Bill Intelligence
AI EXPENSE ENGINE & FINANCIAL LEDGER

ENTERPRISE CLIENT USER GUIDE • V1.0

Client Operating Manual & Workflow Guide

A complete, step-by-step visual handbook for onboarding, uploading bills, split-screen OCR verification, customer/vendor management, bilingual operations, and advanced financial reporting.

⚡ Instant Vision OCR
Extracts line items, dates, vendors, customers, taxes, and totals in seconds.

🌐 Bilingual (EN & اردو)
Authentic Nastaleeq typography with Pakistani B2B dictionary and currency support.

📊 Real-time Auditing
Duplicate invoice guardrails, vendor merging, and multi-horizon analytics.

Confidential & Proprietary • September 2026

1 System Overview & Quick Start

The **AI Bill Intelligence & Expense Platform** transforms paper invoices, digital PDFs, and cash receipts into verified financial ledgers. Whether you manage a clinic, distribution business, retail outlet, or enterprise office, it eliminates manual data entry while maintaining rigorous forensic audit trails.

3-Step Instant Operational Workflow

STEP 01

Upload Bill / Receipt

Drag & drop photos or PDFs into the upload center. Multi-page receipts and batch uploads are supported.

STEP 02

AI OCR Extraction

The vision engine reads line items, quantities, dates, taxes, and totals with spatial bounding boxes.

STEP 03

Verify & Approve

Compare extracted data side-by-side with the original image. One click locks and commits the invoice.

System Navigation Matrix

MODULE	ROUTE / URL	CORE BUSINESS FUNCTION
Executive Dashboard	`/dashboard`	Real-time spending KPIs, monthly trend velocity, category allocations, and top vendors.
Invoices Ledger	`/invoices`	Master financial ledger with status filtering, customer linking, and one-click Excel export.
AI Bill Upload	`/invoices/upload`	Batch document ingestion with auto-preprocessing and duplicate invoice protection.
Manual Bill Entry	`/invoices/new`	Direct recording of offline receipts, fuel chits, and custom supplier vouchers.
Vendors Directory	`/vendors`	Supplier registry tracking total spend, bill count, NTN records, and duplicate merger tools.
Customers Directory	`/customers`	Client management tracking purchaser profiles, phone numbers, and linked invoices.
Reports & Analytics	`/reports`	Multi-horizon analysis (Weekly, Monthly, Yearly, Custom) with single-vendor isolation.
System Settings	`/settings`	User profile, default currency (PKR), language selection, and custom category management.

> 💡 **Ergonomic Navigation Tip**
> The navigation sidebar is anchored to the left in English for consistent desktop workflows. On mobile and tablet devices, tap the **Menu Hamburger** button on the top header to reveal the slide-out navigation drawer.

2 AI Bill Ingestion & Split-Screen Review

CORE PROCESSING

1. Uploading Documents `/invoices/upload`

The upload center processes single or batch invoices. It accommodates high-resolution camera photos from smartphones, scanned flatbed images, and digital PDF bills downloaded from utility providers.

Supported Formats & Protection

- **Image Formats:** JPEG, PNG, WebP (standard phone photos).
- **PDF Documents:** Single or multi-page digital invoices.
- **Duplicate Guardrail:** Automatically alerts operators if an identical file or bill number already exists to prevent duplicate disbursements.

Automated Pipeline Stages

- **Preprocessing:** Automatic contrast leveling and skew straightening.
- **Vision Extraction:** AI parsing of tables, line items, and totals.
- **Financial Math Audit:** Verifies Subtotal + Tax - Discount = Total.
- **Ready for Review:** Queued for operator split-screen verification.

2. Split-Screen Interactive Review Workspace `/invoices/[id]/review`

The review screen pairs the original physical receipt on the left with an editable verification form on the right, ensuring 100% accuracy before ledger commitment.

FEATURE	OPERATION & BUSINESS BENEFIT
Interactive Viewer	Pan, zoom, fit-to-page, and rotate controls allow operators to inspect small print and blurred digits with ease.
Spatial Bounding Boxes	Clicking any input field on the right (e.g. Total, Invoice Date, or Line Item) instantly illuminates an indigo bounding box around the exact source text on the receipt.
Editable Line Items	Add, remove, or modify items, quantities, and rates. The subtotal, tax breakdown, and grand total recalculate dynamically.
Approve & Lock Invoice	Finalizes the invoice, moves status to `Verified`, logs an immutable audit trail, and updates reports.
Re-extract with AI	Allows operators to re-run the vision model with enhanced sensitivity if the initial capture was partially occluded.

> ⚠️ **Audit Safeguard**
> Once an invoice is **Approved**, its status transitions to `Verified` and its financial data is locked to maintain forensic audit integrity for accounting and tax reporting.

3 Financial Ledger & Filtering Engine

Master Invoices Ledger `/invoices`

The Invoices Ledger serves as the central operational hub, detailing every verified bill, manual voucher, and pending document with live running metrics for **Total Records** and **Net Financial Spend** in Pakistani Rupees (PKR).

Universal Search Bar

Instant multi-field search across:

- **Bill Number:** e.g. `INV-84920` or `MAN-49102`
- **Vendor:** Matches English and Urdu names (Metro, LESCO, رائل فینز)
- **Customer:** Matches recipient clients (Noreen Kouser, نورین کوثر)
- **Notes:** Searches internal comments and purchase references

Multi-Attribute Filter Suite

Narrow ledger records precisely:

- **Status Filter:** Verified, Needs Review, Pending, Rejected.
- **Vendor Filter:** Isolate bills from specific suppliers.
- **Category Filter:** Filter by Groceries, Fuel, Utilities, Repairs, etc.
- **Date Windows:** Specify Start Date and End Date range limits.

One-Click Filter Reset & Export Center

When filters are applied, a prominent **Clear Filters / فلٹرز ختم کریں** button appears. Clicking it restores the full ledger instantly. Data can be exported at any time:

EXPORT OPTION	STANDARD FORMAT	PRIMARY PURPOSE
Export CSV	Universal RFC-4180 CSV standard	Direct import into QuickBooks, Xero, Tally, or custom ERP systems.
Export Excel	Formatted Microsoft Excel (`.xlsx`)	Auditor presentations, boardroom reports, and pivot table analysis.

Verification Status Matrix

STATUS BADGE	ENGLISH LABEL	URDU LABEL	OPERATIONAL WORKFLOW STATE
Approved	Verified	منظور شدہ	Bill verified, locked, and fully reflected in executive financial totals.
Review	Needs Review	جائزہ درکار	AI extraction complete; awaiting human operator sign-off.
Pending	Pending	زیر التواء	Ingested into pipeline; background AI OCR actively running.
Failed	Rejected	مسترد شدہ	Document illegible, invalid, or manually flagged by reviewer.

4 Directories & Bilingual Engine (اردو)

1. Vendor Intelligence & Duplicate Merger Tool `/vendors`

The vendor directory aggregates all commercial entities supplying goods or services. It tracks lifetime spending totals, invoice counts, tax NTN credentials, and last transaction dates.

> 🔗 **How to Merge Duplicate Vendors**
> When a supplier appears under multiple aliases (e.g. *"Hafiz Electric Store"* and *"Hafiz Electric Store (Royal Fans)"*), click **"Merge Duplicate Vendors"**. Select the primary target vendor, check the duplicate variations, and click **Merge**. All past bills, line items, and histories will be permanently consolidated under the primary vendor name.

2. Customer Profiles & Client Accounting `/customers`

For businesses that issue bills or manage patient/client expenses, each customer profile tracks lifetime purchases, contact phone numbers, delivery addresses, and itemized invoice histories.

3. High-Fidelity Urdu Localization Engine

The platform features a native bilingual engine calibrated specifically for Pakistani business terminology and Nastaleeq typography.

Bilingual Experience Features

- **One-Click Switcher:** Switch between `EN` and `اردو` instantly from the top header or sidebar.
- **Jameel Noori Nastaleeq:** Authentic ligature-aware Nastaleeq rendering with customized line heights.
- **Pakistani Names Engine:** Translates composite names (e.g. *NOREEN KOUSER* → نورین کوثر, *Hafiz Electric* → حافظ الیکٹرک).
- **Independent Left Navigation:** The sidebar remains rigidly fixed in English on the left, keeping workspace ergonomics natural.

Common Financial Terminology

ENGLISH	URDU EQUIVALENT
Total Spend	کل اخراجات
Verified Bills	تصدیق شدہ بلز
Customer Profile	کسٹمر پروفائل
Clear Filters	فلٹرز ختم کریں
Upload Bill	بل اپلوڈ کریں

5 Reports, Best Practices & FAQ

Financial Intelligence & Comparisons `/reports`

The Reports suite provides macro and micro business intelligence. You can analyze spending momentum across **Weekly**, **Monthly**, **Yearly**, or custom date windows.

Vendor Isolation Matrix: Select any single supplier (e.g. *"LESCO Electricity"*) to view their isolated cost curve, average transaction size, and historical comparison.
Category Allocation: Visualizes where capital is distributed across essential operational buckets (Fuel, Utilities, Supplies, Raw Materials, Inventory).

Best Practices for Flawless AI Extraction

RULE 01

Lighting & Contrast

Capture receipts in bright, even lighting. Avoid heavy shadows across price or quantity columns.

RULE 02

Flat Alignment

Flatten folded slips. Keep the camera parallel to the document to minimize distortion.

RULE 03

High Resolution

Use original camera photos. Avoid sending compressed thumbnails via chat applications.

Frequently Asked Questions (Client FAQ)

QUESTION	ANSWER & RESOLUTION
What if the AI misses a line item?	Click **"+ Add Item"** on the review screen to insert any row manually. Subtotals and grand totals will rebalance immediately.
How do I record offline cash receipts?	Open **"Add Manual Bill"** (`/invoices/new`) to enter vendors, dates, and amounts directly without uploading an image.
Can multiple accountants collaborate?	Yes. All verified records, edits, and audit logs synchronize instantaneously across all connected client devices.
How do I backup our ledger?	Navigate to **Invoices** and click **"Export Excel"** or **"Export CSV"** to download complete ledger snapshots anytime.

✅ **Launch Certification**
This platform has been audited for UI/UX ergonomics, bilingual consistency, mobile responsiveness, and high-performance financial data processing. Ready for immediate enterprise deployment.